XRPL Validator — Daily Engineering Report · 2026-08-12

Period: 2026-08-12 00:00 to 24:00 US Eastern (2026-08-12 04:00 to 2026-08-13 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(xrpl-node): DX_VALCHECK compares EXACTLY, at STAmount precision** (`adca53a` , xrpl-validator) The target is byte parity: a validator serialises the stored decimal into the SLE and hashes it, so one digit apart is a different state hash and a diverged node however small the number. The old check could not certify that — it parsed both sides into f64 and compared against a 1e-12 tolerance.
- **fix(xrpl-ledger): Number division DISCARDS its remainder at the 16-digit scale** (`e7183b3` , xrpl-validator) 129 -> 74 byte-parity hits, 43% of the backlog, from one line. Payment 89 -> 42, OfferCreate 29 -> 21, no key-level regressions.
- **fix(xrpl-ledger): a single-asset withdrawal pays what its adjusted tokens are worth** (`bbf6ee3` , xrpl-validator) AMMWithdraw 7 -> 1 hits. #105816437 AC7713F9 and #105838164 181F8088 go fully clean; the AMM family drops 11 -> 5.
- **fix(xrpl-ledger): port rippled's Newton-iteration square root** (`36a6fdb` , xrpl-validator) AMMDeposit 4 -> 0. #105779597 and #105828322 go fully clean; the AMM family drops 5 -> 1, and only a tfWithdrawAll hit remains.
- **fix(xrpl-ledger): withdraw-all pays two rounded steps, not a fused muldiv** (`81180a1` , xrpl-validator) Closes the last AMM hit. #105796380 2437575D goes AMM-clean; the AMM value family is now ZERO, from 11 this morning.
- **fix(xrpl-ledger): the DIRECT walk charges the input issuer's transfer fee too** (`cf36009` , xrpl-validator) #105877543 435A9AEB -> 0 hits. It is the specimen the direct walk was missing when this was attempted and reverted earlier on 2026-08-11 for want of a failing ledger.
- **fix(xrpl-ledger): multiPath prices a clamped pool fill at the OFFER's quality** (`4a97124` , xrpl-validator) OfferCreate >=1000-ulp cluster 14 -> 5 hits from one rule. #105802230 EF1642A9, #105284280 BE7884AA and #105666830 A8830CA4 go fully clean; #105930662 40FB322E and #105954798 D5887FD7 each drop to one.

- **fix(xrpl-ledger): store offer residuals at STAmount precision, not exactly** (`4556384`, xrpl-validator) `TOffer::consume` subtracts through STAmount, so a maker's remaining side comes back rounded to 16 significant digits (half-even, via `Number` under fixUniversalNumber). `me_sub` is EXACT — it keeps every digit — so whenever `gives0 - give` needed more than 16 digits we stored precision the ledger cannot hold and rippled does not have. Being more precise than rippled is a defect: the validator serialises the stored …
- **fix(xrpl-ledger): charge the input transfer fee once — the walk owns it** (`41ef4ac`, xrpl-validator) `payment.rs` and `offer.rs` both quoted BookStep.cpp:770 and both applied the issuer's rate, and neither comment knew the other existed. `measure_hop` GRANTS a balance and differences it, and the walk debits the fee per fill, so the fee was already inside `consumed` before the reverse pass grossed it up again — the rate compounded.
- **fix(xrpl-ledger): price bridged fills at the filed BookDirectory rate** (`e2a9c99`, xrpl-validator) An offer carries two rates and they are not the same number: the rate it is FILED under (low 64 bits of `BookDirectory`, 16-digit mantissa, fixed at creation) and its current `TakerPays / TakerGets`, which drifts every time a residual is re-rounded. rippled prices a PARTIAL fill at the filed rate and a FULL one at the offer's own amounts. The direct walk has done this since `full_offer`; `cross_bridged` priced both …
- **fix(xrpl-ledger): reprice a rested remainder at the ENCODED ratio** (`070ca31`, xrpl-validator) Two changes to `offer.rs`; only the second moves any ledger.

Independence program

No independence record for UTC day 2026-08-12; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Commit-Reveal Randomness Beacon for Draws** (gaming) Add a new reserve-bearing ltRANDOM_BEACON object (keylet::randomBeacon(account, seq)) created by RandomCommit, which stores sfCommitHash = sha512Half(
- **Multilateral Netting Settlement Batches** (settlement) Add a new ttNETTING_SETTLE transaction that carries a bounded (max 8, mirroring Batch's inner-transaction cap) STArray of pre-signed Payment 'obligati
- **Structured Travel-Rule Payload on Checks** (compliance) Add an optional, size-bounded sfCompliancePayload SField (VL blob, max ~512 bytes) to CheckCreate and CheckCash, carrying FATF originator/beneficiary

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-12 04:00	—	xrpl-validator	`adca53a`	feat(xrpl-node): DX_VALCHECK compares EXACTLY, at STAmount precision
08-12 06:22	—	xrpl-validator	`e7183b3`	fix(xrpl-ledger): Number division DISCARDS its remainder at the 16-digit scale
08-12 13:37	—	xrpl-validator	`bbf6ee3`	fix(xrpl-ledger): a single-asset withdrawal pays what its adjusted tokens are worth
08-12 14:36	—	xrpl-validator	`36a6fdb`	fix(xrpl-ledger): port rippled's Newton-iteration square root
08-12 15:23	—	xrpl-validator	`81180a1`	fix(xrpl-ledger): withdraw-all pays two rounded steps, not a fused muldiv
08-12 16:10	—	xrpl-validator	`cf36009`	fix(xrpl-ledger): the DIRECT walk charges the input issuer's transfer fee too
08-12 18:36	—	xrpl-validator	`4a97124`	fix(xrpl-ledger): multiPath prices a clamped pool fill at the OFFER's quality
08-12 20:58	—	xrpl-validator	`4556384`	fix(xrpl-ledger): store offer residuals at STAmount precision, not exactly
08-12 22:38	—	xrpl-validator	`41ef4ac`	fix(xrpl-ledger): charge the input transfer fee once — the walk owns it
08-12 22:38	—	xrpl-validator	`e2a9c99`	fix(xrpl-ledger): price bridged fills at the filed BookDirectory rate
08-13 03:55	—	xrpl-validator	`070ca31`	fix(xrpl-ledger): reprice a rested remainder at the ENCODED ratio

Times in UTC. A commit appears on the day it was committed to main or merged into it.